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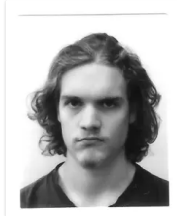
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The GUF (Grand Unified Facts) of Reality

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On The GUF (Grand Unified Facts) of Reality: towards a simple abstraction



Author of this Thesis and Dissertation: Waldo Pelser

Abstract:

All of science and even large parts of modern mathematics is reduced or made redundant by a complete, consistent and perfectly expressive knowledge base that proves Linear Space $\equiv$ Non-Linear Space and expresses reality as a multidimensional Euclidean Space and particles and radiation as infinite-metrics and projections from them respectively. All natural phenomena from plasma discharge to superimposed entangled waves of consciousness are explained as interactions of these metrics. All relevant unsolved mathematical and scientific problem from Hilberts Sixth Problem to the P vs. NP millennium problem is solved with tautological propositions. A pre-Atlantian and pre-Deluge nature of earth's atmosphere is discussed with relevance to the authors theological views as this is a comprehensive dissertation with theological, scientific and mathematical implications.

The Problem of Doctrine: A mathematical solution

Science is an **empirical** paradigm of theory and is thus insufficient for solving the problem of defining real phenomena according to **first-order logic (first-order predicate calculus of quantified variables)** – the Scientific Method discovered the early axioms of geometry used in surveyance of 3D space where motion is only treated empirically with equations of motion but not axiomatically leading to a separation of knowledge into maths and science. It cannot be expected that a solution to **Hilbert's Sixth Problem** would not potentially **trivialize** science and certain mathematics completely. The **abstraction of new axioms** in Euclidean Geometry give **well behaved** solutions to Hilbert's Sixth Problem defining all of reality.

"Hilbert's sixth problem is to axiomatize those branches of physics in which mathematics is prevalent. It

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Yves Couder's experiments are one example of how the microcosm works like the macrocosm debunking the Copenhagen interpretation of quantum mechanics, which most of the dogma such as quantum chaos theory is based on, and that the forces that make up the atom are the same as those that make up the planets, stars and galaxies... [<http://www.youtube.com/watch?v=H9y0mJd9p0c>]

all of which Carl Sagan has said long before... [http://www.youtube.com/watch?v=47n7*ym9b04]

Science has not solved Hilbert's Sixth Problem and is thus FUNDAMENTALLY FLAWED!... [http://en.wikipedia.org/wiki/Hilbert%27s_sixth_problem]

Units like Meters and Teslas are replaceable and only the ratios between them form the scientific constants and 'Hilbert's Sixth Problem' is solved when the constants of 'Science' are shown to be 'Mathematical'... [<https://www.youtube.com/watch?v=xuUd8VLA1s>]

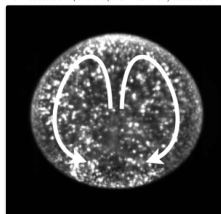
Many "established" scientists claim that God does not exist yet they admit they where wrong about how a bar-magnet works... [<http://phys.org/news/18458077.html>]

Atoms and all particles have been found to have a **Hoplon** structure... [<http://www.sciencemag.org/cgi/doi/10.1126/science.1261314>]

Galaxies, Stars and Planets have also been found to have a **Hoplon** structure... [<https://doi.google.com/10.1038/nature12614>]

Understanding Euclidean Geometry in higher dimensions is required... [http://en.wikipedia.org/wiki/10d_geometry]

Even Yves Couder drops before quantum mechanically because of their **Hoplon** structure asseen on page 34 of his thesis



"Figure 2.5: A snapshot of a source (at a maximum of trajectory) of a 5-cm silicone oil droplet on a high viscous bath that oscillates at 50 Hz and at an acceleration $a = 3-5$. The internal flows are revealed by fluorescent microphotographs."

[<https://doi.org/10.1038/47010a>]

Abstract

Following Rutherford's 1920 historical hypothesis of the neutron as a compressed hydrogen atom in the core of stars, the laboratory synthesis of the neutron from protons and electrons was claimed in the late 1960 by the Italian priest-physicist Don Carlo Borchgrevink and his associates via a metal chamber containing a partially ionized hydrogen gas at a fraction of 1bar pressure traversed by an electric arc with 5J energy and microwaves with 1010s-1 frequency. The experiment remained unverified for decades due to the lack of theoretical understanding of the results. In this note we report various measurements showing that, under certain conditions, electric arcs within a hydrogen gas produce neutral, hadron-size entities that are absorbed by stable nuclei and subsequently result in the release of detectable neutrons, thus confirming Don Borchgrevink's experiment. The possibility that said entities are neutrons is discussed jointly with other alternatives. Due to their simplicity, a primary scope of this note is to stimulate the independent re-run of the tests as conducted or in suitable alternative forms. [<http://www.hopfon.com/donborchrevink/neutron.htm>]

Neutrons and Protons are fictitious constructs to explain mass and charge yet **Hopfonic** structures have mass and charge simply due to their conservative motion forming coherent structures in the aether (medium in which they are formed) which itself consists of smaller atoms of different parity and chirality... [<http://www.hopfon.com/>]

As in the old literature of Dmitri Ivanovich Mendeleev, atoms where things with charge and mass thus it's clear that when they changed the sentence from "the electron of charge" to "the charge of an electron" they made the mistake of thinking it's "one electron for each bond" when it's "one bond for each charge"... [http://en.wikipedia.org/wiki/Feynman's_charges]

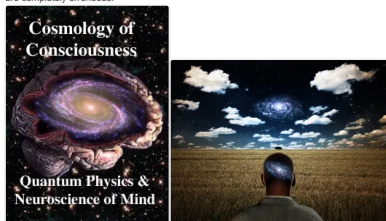
If Lord Kelvin, who proposed in 1867 that atoms could be knotted tubes of Aether, is wrong according to QM then the best answer to the structure of a particle cannot be determined by it... [<http://hopfon.com/>] [<http://www.youtube.com/watch?v=evrj7C0P96g>]

On the Standard Models: Toward a Mathematical Context

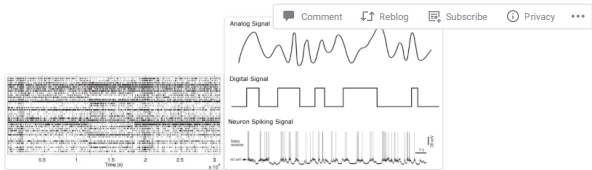
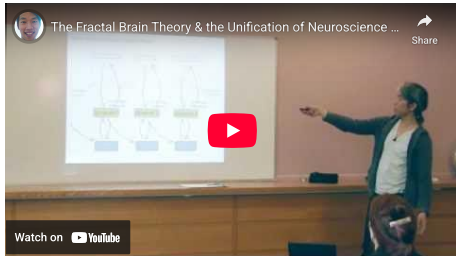
Propaganda of Atheological Dogma:

'Theoretical Physics' is one of the 'Atheological' departments at university that have no evidence what-so-ever for their theories and thus should be named 'Hypothetical Physics' for their wild speculations since a degree in the field is as scientifically valid as a university 'Theology' degree. The only difference between the two is that the former is propagated by the conglomerate mainstream media.

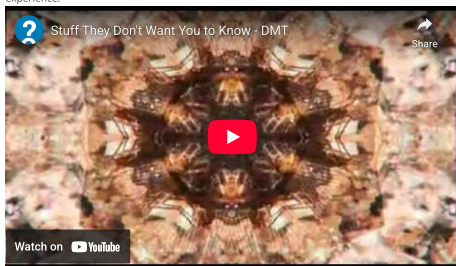
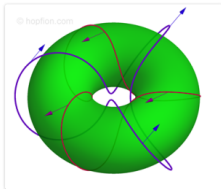
These "scientists" claim 'Higgs Boson' is derived purely from mathematics yet 'Hilbert's Sixth Problem' shows this to be a fallacy as it has never been solved by 'Supersymetry Theory' and thus their assumptions are completely erroneous.



Consciousness is already explainable through electrical physics not quantum dogma and the countless attempts of QM to worm it's way into neuroscience with elaborate hypotheses has failed yet somehow their ideas persist through the literature where the electrical universe models clearly show that consciousness is the sum of all electrochemical activity in the brain creating a Network Topology that uses synchronous all-or-nothing conditions in combination with analogue impulses creating a Analogue-Digital computer Hybrid. Consciousness is the result of superimposed electro-chemical waves that are each a consequence and cause of the fractal antenna design of the brain where each neuron and aggregates of neurons act as transmitters and receivers of a specific waveform that propagates through the brain forming the superimposed coherent wave which is the mind.



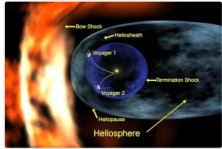
Antimatter and all other subatomic particles are very vague short lived observation in particle physics yet at CERN they attempt to build a model of coherent debris in the medium after high speed collisions of **Hopfions** [<http://hopfon.com/>] i.e.w. they smash matter into smaller matter then attempt to fit the pieces together in a model to try and prove the already debunked Big Bang Hypothesis thus this new approach to the explanation of reality is a far-cry from any standard model and quantification paradigm of science. Author's favorite videos about conscious experience:





The Cosmic Plasma Orb or Sun:

The magnetic forces on the sun are 10,000,000 times that of its gravitational force and are responsible for the Termination Shock that protects the solar system from the galactic currents of UHECPs.



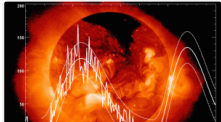
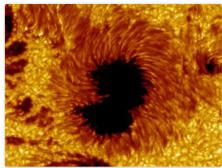
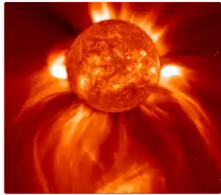
[<http://homsaastroblog.com/images/2010/01/Voyager-at-Termination-Shock1.jpg>]

Gravity is not the governing force in the universe so the assumptions made by cosmology in the Standard Model of the sun are based on the myth that gravity is causing the sun to collapse on itself with only nuclear fusion to keep it from imploding into a white dwarf or black hole and that the sun will one day run out of fuel. The sun is not a isolated body that must consume itself over it's lifetime but instead an electric plasma orb that is connected to the rest of the galaxy through a network of currents.

The sun's corona [<http://en.wikipedia.org/wiki/Corona>] is unexplainable through the Standard Model (Fusion Model). Coronal Heating is another unexplainable phenomena where the surface of the sun (photosphere) has an **apparent** temperature of 6000K but the corona (chromosphere) has an **apparent** temperature of 2 000 000K so the problem is proving how the sun could heat the corona to millions of degrees without heating the surface yet this is not a problem in the electric model where the sun receives it's energy from **outside** where the corona is the first place this energy would be expected to express itself and here particle **acceleration** occurs.

[<http://www.plasma-universe.com/images/0/0e/Sun-corona.jpg>]

The fact that particles in the solar wind accelerate (increase velocity) means that the sun has an enormous electric field. Another unexplainable phenomenon in the fusion model are Sun Spots and the important observation to be made is that the sunspots are where the plasma filament that follow the magnetic lines of force emerge from the surface leaving a dark centre (penumbra) and if the sun is radiating energy from its core outward into space then it should be brightest there. The penumbra is where we can see deepest into the sun and that is where it is the coldest



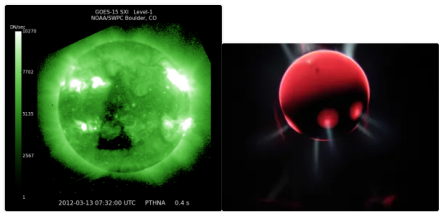
[http://www.oneminuteastronomer.com/wp-content/uploads/2009/09/Sunspot-group_1.jpg]

[http://cache.io9.com/assets/images/8/2011/08/xlarge_ssn_predict_1.gif]

"The biggest discovery of all would be the inner workings of the solar dynamo. Deep beneath the sun's visible surface, massive currents of electrically-charged gas (plasma) flow in a cycle that creates the sun's powerful magnetic field. Almost all solar activity from sunspots to solar flares is driven by the inner dynamo.

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"Understanding how the dynamo works is a holy grail for stellar physics," says Pesnell. "It is the key to forecasting solar activity and space weather." [http://vanshardware.com/wp-content/uploads/2012/03/latest_sst_triangle_hole.png]



[http://tsdoisgo.blogspot.com/2009_08_01_archive.html]

Astrophysicists say the fusion model has been tested and verified but this is a lie. Although each of the steps in the hydrogen to helium fusion reaction – Proton-Proton Chain Reaction – has been verified experimentally, the overall reaction has never been achieved or verified. All of the suns processes have been proven to be electrical. There are a number of phenomena that have Ad-Hoc explanations but no explanation in the Standard Model:

Heavy Elements,

Solar Spectrum,

Neutrino deficiency,

Neutrino Variability,

Solar Atmosphere,

Differential Rotation by Depth,

Equatorial Plasma torus,

Sun spots,

Sun Spot Migration,

Sun Spot Penumbra,

Sun Spot Cycle itself,

Magnetic Field Strength,

The Even Magnetic Field,

Helio Seismology, and

Solar Density and Changing Size.

<http://www.thunderbolts.info/wp-content/uploads/2012/11/Sunset-550x317.jpg>



All these phenomena are explainable by Plasma Cosmology.

Special Relativity:

Einstein's theory of SR (Special Relativity) and QM (Quantum Mechanics) do not reconcile because according to QM particle motion is uncertain (Heisenberg's Uncertainty Principle) which applies to both particles and also to empty space. Over very short intervals phantom/virtual particles can appear where the shorter the time of the appearance, the more massive the particles can be. At very short intervals, virtual particles are massive enough to form 'black holes'. One cannot extrapolate simultaneously both quantum mechanics and general relativity to minute distances as the theories diverge.

QFT (quantum field theory) combines SR and QM in a problematic structure where Practical experiments have tested the questionable aspects of QFT using Bell's inequality. The predictions of QFT, not the mechanism that generates those predictions, are relativistic. Motion in space-time is impossible since any

four-vector (4-Force, 4-Momentum, 4-Velocity) is self-referential $\frac{dt}{dt}=1$ as the t-coordinate = 1 in $(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}, 1)$, hence everything that is said to occur with the 'arrow of time' does so at time=1 all the time.

One Sentence Proof #1:

"Moving in space-time is impossible because it requires motion in time and motion in time is self-referential." [http://www.rebelscience.org/Crackpots/notorious.htm]

One Sentence Proof #2:

"Moving in space-time is impossible because an evolution parameter (time) cannot be its own evolution parameter." [http://www.rebelscience.org/Crackpots/notorious.htm]

GPS systems supposedly use GR(General Relativity) for their calculations but proper-time variable τ (tau) is a third-reference frame and according to SR there can only be two-references frames leading to a contradiction between the mathematics and theory of SR.

Mass-energy equivalence:

Tesla was right to say $[E]=(m)[c]^4/2$ is "mathematical poppy cock" as the equation can be deduced from the resultant motion of a body that has equal tangential (centrifugal) acceleration as it has radial (toward centre of motion) acceleration resulting in a $\sqrt{2}$ velocity using Pythagoras of $\{45\}^\circ$ triangle and substituting back into classical Kinetic Energy $[E]_K=(1/2)(m)(v)^2=(1/2)(m)(\sqrt{2}(c))^2=(1/2)(m)(2)(c)^2=(m)(c)^2$ Motion is tangentially and radially equal at the equator of a **Hoplon** where the forces between the sink and source poles are at equilibrium causing Matter(**Hopfions**) to collect on the equatorial plane.

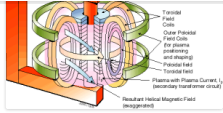
[<http://www.softcom.net/users/greebo/sincklen.htm>]

Regardless of the simple geometric interpretation of $[E]=(m)(c)^2$, it is flawed and has no application in **fusion** and **fission** nuclear reactions! **Fission** weapons tests have found ice build up in the mushroom clouds proving radiant energy is being compensated for with the absorption of ambient heat.

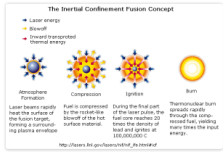


Fusion reactors have never been able to "break-even" (reach energy output that equals energy input) such as "Magnetic Confinement Fusion" reactors

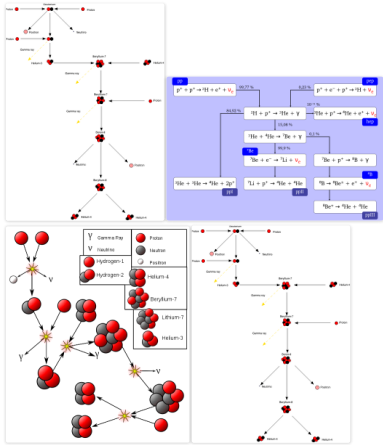




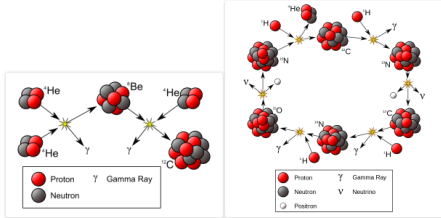
and 'Inertial Confinement Fusion' reactors.



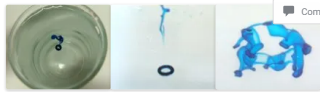
Fusion 'Nuclear Chain Reactions' (NCR) such as the 'proton-proton' chain reaction



and Carbon-Nitrogen-Oxygen (CNO) cycle has never been verified in any way to exist.



"Atoms of the same element can have different numbers of neutrons; the different possible versions of each element are called **isotopes**", however, neutrons are not necessary to explain atomic mass as it is a result of **hoplon** flux density where **hoplon** harmonics determine the quantification of mass (neutron).



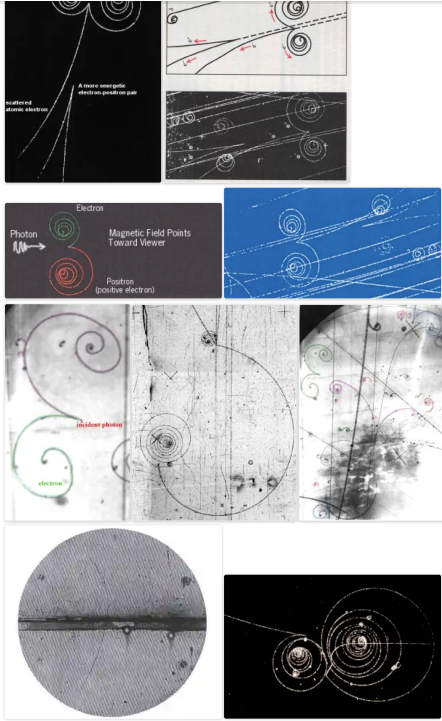
The statements "Antineutrinos are the antiparticles of *neutrinos*, which are neutral particles produced in nuclear beta decay. The antineutrinos observed so far all have right-handed helicity (i.e. only one of the two possible spin states has ever been seen), while the neutrinos are left-handed... Antineutrinos, like neutrinos, interact with other matter only through the gravitational and weak forces, making them very difficult to detect experimentally. Neutrino oscillation experiments indicate that antineutrinos have mass, but beta decay experiments constrain that mass to be very small." [Wikipedia] suggests the neutrino quanta are **hopfions**.



It is assumed that a positron-neutrino pair that annihilates with an electron-antineutrino pair is emitted by beta decay cause of gamma ray detection (annihilation of e^+ and e^-) but "what happens in neutrino antineutrino annihilation? The two particles meet at a single point and annihilate each other, producing a virtual Z boson, which is the neutral (i.e. no electric charge) carrier of the weak nuclear force. This Z boson then immediately decays to produce another particle/antiparticle pair, either a new pair of neutrinos, two charged leptons, or a quark/antiquark pair. What you can produce depends on how much energy there is from the colliding neutrinos." "If the antiparticle to a photon is a photon, why doesn't it annihilate itself? A photon has spin = 1. You can't have and conserve spin. And it can't go to a charge object due to charge conservation.

[This is a contradictory point deduced from theory]"

Pair-Production assumes an incident photon to cause electron-positron production when it collides with an atomic nucleus yet equivalently one can assume the gas molecule to ionise and split into positive and negative ions creating the same cloud trails yet a **hoplon** can produce two smaller hopfions at its poles each with opposite density gradients (charge) due to a sudden fluctuation in its own density when coupling with the photon.

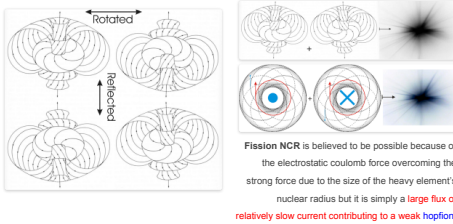


"Cloud chamber photograph by C. D. Anderson of the first positron ever identified. A 6 mm lead plate separates the upper half of the chamber from the lower half. The positron must have come from below since the upper track is bent more strongly in the magnetic field indicating a lower energy."

"Invented in 1952, A bubble chamber is a vessel filled with a superheated transparent liquid (most often liquid hydrogen) used to detect electrically charged particles moving through it. Cloud chambers work on the same principles as bubble chambers, only they are based on supersaturated vapour rather than superheated liquid."

Nevertheless the right handed or left handed toroidal flux (Stereographically Projected Hypersphere Spin) of **hopfions** correspond to matter or antimatter respectively.

Profile of symmetries:



Fission NCR is believed to be possible because of the electrostatic coulomb force overcoming the strong force due to the size of the heavy element's nuclear radius but it is simply a **large flux of relatively slow current contributing to a weak hopfion**.

Besides the fact that Einstein found GR in his dreams, "he tried to interpret a simple equation with two contradictory and paradoxical ideas of mass and energy. In the general interpretation, $[E]_0[k] = [M](C)^2$ defines the relationship between mass and kinetic energy. This means that when a body of mass is decelerated it loses mass and energy and when it is accelerated it gains mass and energy. This mass increase/decrease for all matter is proportional to each body's kinetic energy relative to a **common position of rest** for all matter. In this interpretation of $[E] = [M](C)^2$, energy and mass coexist together. The kinetic energy of an object contributes to its overall mass. Einstein refused to believe in deceleration as a meaningful concept that could be differentiated from acceleration to establish a position of absolute rest because the mass changes caused by motion cannot be measured locally.

The primary meaning of $[E] = [M](C)^2$ is that it defines the mass of the photon as the truest measure of mass. From convention and with no experimental verification, Einstein arbitrarily declared the photon to be a massless particle using Planck's Constant to make the transformation between the mass of an atom and the energy of a massless photon thereby failing to divide Planck's constant into its component parts $(h) = [M](\lambda)$ (C) (mass of a photon times its wavelength times the speed of light).

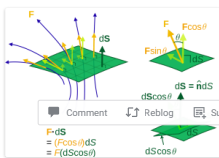
Ignoring the mass of the photon, completely changes the nature of $[E] = [M](C)^2$ from the previous interpretation. In this case, matter still contains kinetic energy that has mass, with the exception that the kinetic energy of the photon has no mass.

If Einstein would have allowed the photon its fair share of the mass, then there would be no case where mass is converted into energy yet then SR would fail for recognising the Aether. Mass has energy that can be measured and energy has mass that can be weighed. Mass and energy are two sides of a coin and can't be separated into the mass of matter and the energy of photons. Mass and energy are the two primary parameters of both matter and photons. One cannot exist without the other. There is no such thing as the long cherished metaphysical idea "pure energy". There is only pure "mass-energy".

Photons can only be produced by the annihilation of matter and antimatter and the universe contains no antimatter that could be combined with its matter to produce energy except in the high energy collision of particles.

The process of annihilation between a positron and electron is very short lived and can only produce photons with a total energy equal to hydrogen's ionization energy of 13 electron volts. In contrast, the mass-energy of a Hopfion is about one billion eV. The Hopfion loses equal amounts of mass to the emitted photons as it drops down into the ground state where the process stops and the atom becomes stable. Einstein failed to understand that all photons travel at $[C]$ through the same inertial reference frame and not just relative to observers. Einstein made the speed of light relative to the observer's frame. It is true, as Einstein claimed, that all observers will measure the speed of light to be $[c]$ in any frame. However this does not mean that they are all measuring the same exact quantity. Capital $[C]$ is the velocity of photons relative to photon rest and small $[c] = [C] - [v]$ is the speed that photons travel relative to moving observers. Due to mass changes in the moving observer's measuring devices, $[c] = [C] - [v]$ always come out to have the same value.

Einstein made this assumption of relative motion not because he could prove it but because no one disproved it. The very nature of the accelerometer and the Doppler effect make it impossible to measure the difference



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The process of annihilation between a positron and electron is very short lived and can only produce photons with a total energy equal to hydrogen's ionization energy of 13 electron volts. In contrast, the mass-energy of a Hopfion is about one billion eV. The Hopfion loses equal amounts of mass to the emitted photons as it drops down into the ground state where the process stops and the atom becomes stable. Einstein failed to understand that all photons travel at $[C]$ through the same inertial reference frame and not just relative to observers. Einstein made the speed of light relative to the observer's frame. It is true, as Einstein claimed, that all observers will measure the speed of light to be $[c]$ in any frame. However this does not mean that they are all measuring the same exact quantity. Capital $[C]$ is the velocity of photons relative to photon rest and small $[c] = [C] - [v]$ is the speed that photons travel relative to moving observers. Due to mass changes in the moving observer's measuring devices, $[c] = [C] - [v]$ always come out to have the same value.

Einstein made this assumption of relative motion not because he could prove it but because no one disproved it. The very nature of the accelerometer and the Doppler effect make it impossible to measure the difference

Einstein's idea of relative motion for matter cannot be reconciled with the absolute motion of photons. If photons move with absolute motion, then the motion of matter must also be absolute. $(M)=(E\over{[C]^2})^{(2)}$ defines a body of matter's excess mass associated with its absolute motion through rest. When a body of matter is accelerated to any velocity (v) relative to this frame, its mass increases with its kinetic energy $[E_k]=[M][C]^2$. At a velocity of about 86% of the speed of light, a body's mass is doubled with a kinetic mass that is equal to its rest mass.

We can measure our own absolute motion to be approximately 375 km/sec in the general direction of the constellation Leo. This increases the total mass of our individual bodies by only about 1/4 gram. This increase in mass might seem small until we realize that this amount of mass represents as much kinetic energy as an atomic bomb. Even though this enormous kinetic energy of our bodies is real, there is no way to tap or even detect the several kilotons of kinetic energy contained in the absolute motion of everyone's individual body."

Edited [<http://www.crdm.com/living-universe/025-how-einstein-was-wrong-about-E=MC2.html>]
Mass in motion creates Force and Energy thus $(F)=[m][a]$ can be called **mass-force equivalence**.

Sonoluminescence:

It is assumed that **spectral measurements** of Sonoluminescence confirms a hypothesis that the bubbles have a **hot plasma core** of about, yet not agreed upon, 20000 degrees Kelvin – 25000 degrees Fahrenheit – (four times hotter than the surface of our Nuclear Sun Model) when compressed with sound waves above 18000 cycles per second but **spectral measurements also confirms a hypothesis that it has a cold plasma core!**

These equations were made using **five major assumptions**, with four of them being common to all the equations:

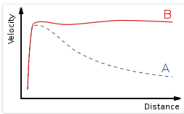
- 1. The bubble remains spherical.
- 2. The bubble contents obey the ideal gas law.
- 3. The internal pressure remains uniform throughout the bubble.
- 4. No evaporation or condensation occurs inside the bubble.

The fifth assumption, which changes between each formulation, pertains to the thermodynamic behavior of the liquid surrounding the bubble. These assumptions severely limit the models when the pulsations are large and the wall velocities reach the speed of sound.

All these **assumptions** have been proven to be wrong in cavitation experiments. As the bubble collapses it creates a nano-vortex which rips the ions from their bonds in a cold plasma.

Astrological Astronomy:

Mainstream Cosmology teaches us that 'Dark Matter' exists in order to explain the non-inverse square proportionality that is observed under the Einstein Model yet Dark Matter has never been proven to exist and the same is true for Neutron Stars (used to explain Pulsars) and Black Holes (crucial for proving the Big Bang Hypothesis).



"Rotation curve of a typical spiral galaxy: predicted (A) and observed (B). Dark matter can explain the 'flat' appearance of the velocity curve out to a large radius."

[<http://en.wikipedia.org/wiki/File:GalacticRotation2.svg>]

I am aware of many EM(electromagnetic) models such as 'Thunderbolts Plasma Cosmology' that is proven and explains all cosmological phenomena without the need for assuming the existence of abstract celestial objects that have never been discovered.



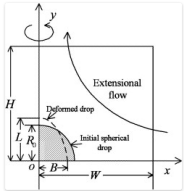
[<http://www.thunderbolts.info/en/>]

This model is improved by the fact that light is not a particle... Comment Reblog Subscribe Privacy ...

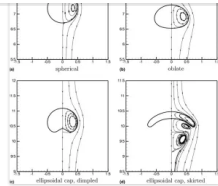
Diffraction (wave-like), Interference (wave-like), Refraction (wave-like), Reflection (wave-like) ect. yet for only one phenomena called 'The Photoelectric Effect' it is assumed that light is also a particle called a 'photon'.

The 'Compton effect'(invented to explain the phenomena) states that the massless photon interacts with an electron to impart energy on it yet if $(F)=[m][a]$ then $(F)=0([m]=0)$ and $(E)=(1\over{2})[m][v]^2([v]=0)$ showing light is obviously not a particle.

This model is further improved by Yves Couder's experiment with silicone drops termed 'walkers' that completely invalidates the Copenhagen Interpretation of QM which claims the nature of the universe is random and probabilistic leaving 'Quantum Chaos Theory' and the 'Uncertainty Principle' obsolete. Yves Couder's experiments proves the forces of the microcosm are the same in the macrocosm giving way to local evidence that CM (Continuum Mechanics) explains the atomic world thus the 'Uncertainty Principle' is instead better explained by 'Fractal Unpredictability' and the normalization of the 'Hydrogen Wave Function' for defining probability density with associated 'Quantum Numbers' are unnecessary thereby proving the atom is better represented as a **Hopfion**.

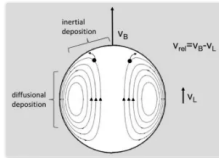


"Fig. 3. Schematic of the computational domain for transient drop deformation after abrupt inception of a uniaxial extensional flow. The flow is axisymmetric with respect to the y axis and the domain lies in the meridional plane. The drop size is exaggerated; the actual domain size has $H = W = 10R_0$."

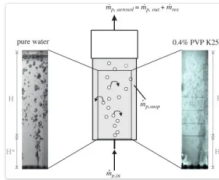


"Fig. 6. Computed drop shapes and the flow field in the final steady state. The streamlines are in a reference frame attached to the centre of mass of the drop. A mirror image has been added to show the full shape of the drops. (a), (b), (c) and (d) correspond to case A, B, D and E in Table 2."

[<http://www.sciencedirect.com/science/article/pii/S0021999106001392>]

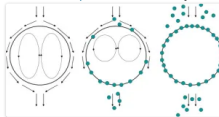


"Fig. 1. Internal circulation (Hill's vortex) inside a rising gas bubble."



"Fig. 3. Mass balance of the bubble column (middle) and pictures taken with a high-speed camera for pure water (left) and for a PVP-water solution (right)."

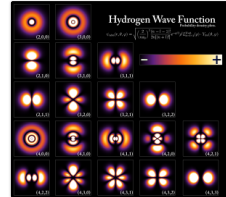
[<http://www.sciencedirect.com/science/article/pii/S0021850212001048>]



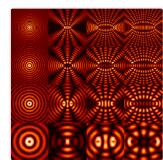
"Fig. 1. Schematic overview of the effect of contaminants on the drag force. From left to right: ultra-pure liquid with free-slip boundary condition at the bubble interface; slightly contaminated liquid with a limited circulation inside the bubble and a stagnant cap at the bottom; fully contaminated bubble, without any internal circulation (full no-slip boundary)."

[<http://www.sciencedirect.com/science/article/pii/S0009250909007362>]

Schrodinger's Equation of the atom is a Laplacian equation that explains only the interaction of light with the atomic ion and not position of the electron nor chemical bonding by Valence shell electron pair repulsion (VSEPR) theory, valence bond (VB) theory or molecular orbital (MO) theory whom contradict each other, however, the 3-Sphere Harmonics are also sufficient.

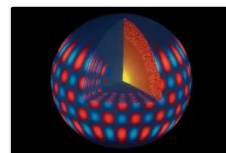


Yves Couder's 'walkers' show the 3-Spherical Harmonics at the surface like longitudinal waves become transverse at the surface of the water pool – as shown on page 30 of 'Bouncing Droplets' the role of deformations – in combination with the Toroidal Flow inside the Hopfionic structure. Harmonics are standing waves caused by interference between oscillations.

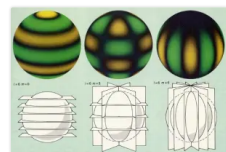


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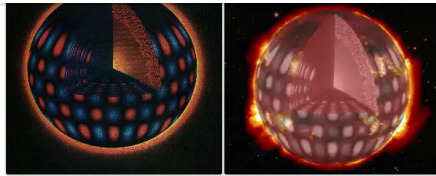
Spherical Harmonics of the sphere and Vector Spherical Harmonics (VSH) of the ball contained by it are both found in Helioselsmology.



"Cut-away view of the Sun showing p and g modes with an accompanying l vs. v power spectrum"

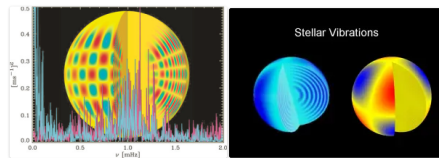


"3 Surface Harmonics at l=6 m=0, l=6 m=3 and l=6 m=6."



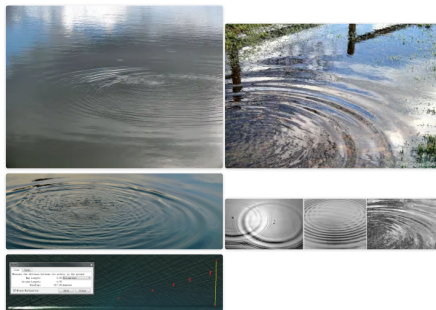
"Explanation: Our Sun is in a continual state of oscillation. Large patches of the Sun vibrate in and out, back and forth, even as the Sun rotates. One mode of Solar oscillation is depicted graphically above, with blue indicating outward motion, and red indicating inward motion. Although sensitive optical solar observatories can only directly detect surface motions, they give information about vibrations occurring much deeper in the Sun. In helioseismology, these oscillations are being analysed and are revealing unprecedented information about the density, temperature, motion, and chemical composition of the entire Sun. "

[National Solar Observatory / Global Oscillation Network Group]



"Spherical harmonics of a standing wave in the Sun. The cut-out on the right-hand side reveals the pattern of the standing wave inside the Sun." [http://homepage.univie.ac.at/guenter.houdek/helios.html]. [Google Images: Helioseismology]

Another model called 'The Dynamic Universe Model' that provides a solution to the 'Newton Three Body Problem' explains that **BlueShift** is as possible in the Einstein Model as **RedShift** yet only the latter is realized.



Blueshifts is seen as much as **redshift** except at sufficiently far enough distances where light moving through more 'Space-Medium' loses energy like water wakes **redshift** in a stagnant pool yet **blueshift** should not occur according to the Cosmological Principle and Space has no medium according to SR. The universe is clearly displaying a type of 'Brownian Motion'. If we were to believe in universal expansion then we must also believe our Milky Way Galaxy is the centre of the universe and of concentric spheres of galaxies at quantized distances according to Hubble's data but this is ignoring **blueshift** and is as erroneous as believing that the Earth is flat and trying to look for its ends or that the Sun revolves around Rome for anywhere one moves about the void it would seem as if you are at it's centre.

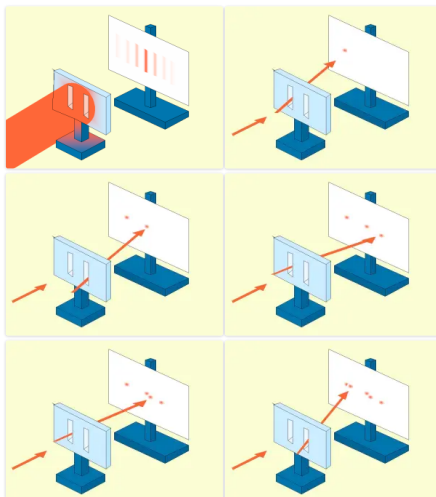
Since there is a third reference frame called 'Aether' the Big Bang Hypothesis becomes disproved as the 'Relativistic' requirement that there are only two is important to the expansion logic in the way that it leaves out the space where in space was to expand which is mathematically self-referential and thus cancels the whole mechanism leaving the entire process redundant.

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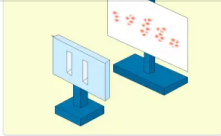
On "Dr. Quantum": Toward the Interpretation of QM

[<http://www.youtube.com/watch?v=Q1YogPAtzhe>]

Double-slit experiment:



changing the transference function(observer effect). There is no one "electron particle" that splits and interferes with itself when it reaches the double slit. Aether goes through both slits but the wake created by aether particles in the cloud allow them to diffract since it was not one electron that they shot it was one charge that is filled with more electrons as in the old literature they referred to an electron of charge not a charge of an electron. Atoms don't release an electron per bond charge but rather a bond charge of electrons as one bond is not one electron but a ratio of one charge to many electrons (aether particle cloud). Charge build up on the tip of the cathode accumulates before being released hence "one electron at a time" is not being released.



The idea that dimensions are separated domains of reality is a speculative fallacy. Dimensions are mathematical tools to analyse physical systems not realms we live in. All dimensions(infinitely many) in a mathematical analysis are required to understand the full nature of a physical system. Dimensions are only components of the physical universe that together make a more realistic image from their kernel about the universe. There are no such things as extra-Dimensions, parallel-Worlds and Multi-verses. Time is a result of the 4D mechanisms that produces it hence it is only a consequent perception of real mechanical motion in a Hopf Fibration with sink and source flux symmetry and self similar and scale invariant topology. Simply put its like a doughnut consisting of very small doughnuts that experience a 1000 cycles before the big doughnut experiences one cycle hence time is a result of fractal processes. Space is not empty otherwise it would be an infinitesimal point yet the Aether medium fills all space. The homogeneous frictionless Aether is an infinitely large fractal domain of conservative motion inside conservative motion of particles making up the fields of particles explaining the 4D mechanism that creates the time dilation and joins the other 3D analytical devices of measurement. Atoms are one of these Hyper-spherical constructs. Even Mendeleev believed in elements and Maxwells 1861-2 paper "On Physical Lines of Force" discusses the concept of vorticity motion as a continuum mechanical explanation of Aether.

<http://www.eric-giblin.com/>
<http://www.fractaluniverse.ac/pape.co.uk/ElectroFractalUniversePublication.pdf>
Stochastic mathematics (probability and statistics) reveal trends not mechanisms hence the name "quantum mechanics" (QM) is an oxymoron that must be replaced with "quantum trends" (QT).

The only mechanical part of QM is the electrostatic nuclear model of the atom which is adopted from electrostatics in "classical mechanics" (CM) thus QM must abandon its classical atomic model.

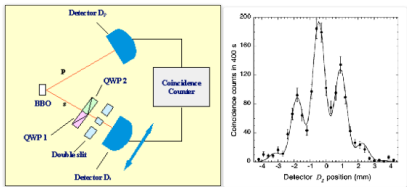
The Classical explains the mechanisms of reality therefore it is called mechanics

The Quantum explains the trends of reality therefore it is to be called trends.

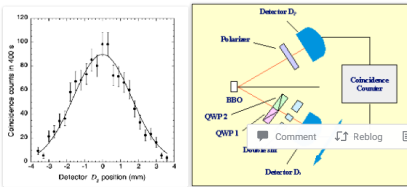
Quantum eraser: <http://grad.physics.sunysb.edu/~amarchi/>



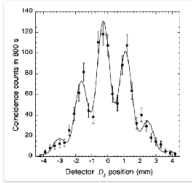
"One fire dot # One photon detection" hence light is only a wave explaining the vectorial photoelectric effect meaning the entire laser beam is entangled explaining how a "photon" absorption into the detector does not effect it's partner unless photon-electron entanglement is possible. The coincidence detector only registers information of one detector that coincides with information on the other thereby revealing a interference pattern.



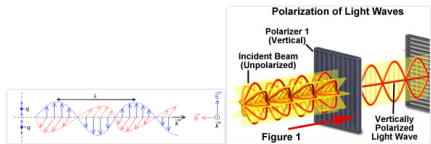
Placing a "quarter wave plate" (QWP) in front of each slit changing linearly polarized light into circularly polarized light where one QWP would change one to right circular polarization while the other would change it to left circular polarization. Coincidence information of light passing through each QWPs, and thus each slit, at one detector coincides with the respective information of light detected at the other thereby losing the interference pattern. Since the information being detected at 'y' coincides only with light moving through one of the slits at 'x'.



Polarizing one of the split beams causes both beams to be so polarized. Polarizing the beam at 'y' to be both 'x' and 'y' polarized causes the same light to move through both slits at 'x'. Information of 'x' and 'y' polarized light detected at 'y' coincides with information of 'left' or 'right' polarized light detected at 'x'. Since the information being detected at 'y' coincides with light moving through both of the slits at 'x'.

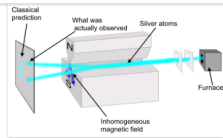


Polarization of light breaks the laws of electromagnetic waves since linear polarization of light causes the light to become an electric, magnetic or electromagnetic wave that has parallel magnetic or electric fields hence polarization proves magnetic and electric fields are components of aether motion. Magnetic or Electric fields cannot exist by themselves if they are mutually induced thus the phenomena is evidence contrary to the transverse EM-wave theory.



Delayed choice quantum eraser:

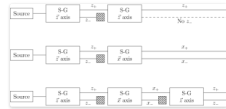
The simultaneous creation of light beams or splitting of one seems to "entangle" them all proving "photons" are merely quanta of light absorbed or emitted by an atom per harmonic cycle and thus not only a indistinguishable QM particle (boson) but rather longitudinal waves in the aether medium.



The Stern-Gerlach Experiment with silver atoms show them to have a magnetic moment not their electrons (such would be an assumption/conjecture) as the successive splitting of the beams prove that atoms orientate themselves during their flight path. The extremely small size of atoms suggest they are well capable of acquiring enough angular momentum during their flight path to have quickly orientated themselves either up or down meaning the "1/2-spin state of electrons" is rather 50% probability of becoming North/South orientated satisfying a classical prediction.

Sequential experiments.

"If we link multiple Stern-Gerlach apparatuses, we can clearly see that they do not act as simple selectors, but alter the states observed (as in light polarization) according to quantum mechanical law."



"Note that the experiment was performed several years before Uhlenbeck and Goudsmit formulated their hypothesis of the existence of the electron spin. Even though the result of the Stern-Gerlach experiment has later turned out to be in agreement with the predictions of quantum mechanics for a spin-1/2 particle, the experiment should be seen as a corroboration of the Bohr-Sommerfeld theory"

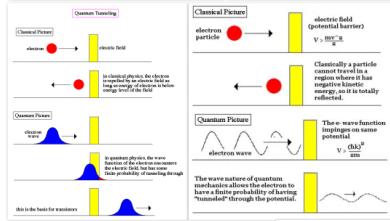
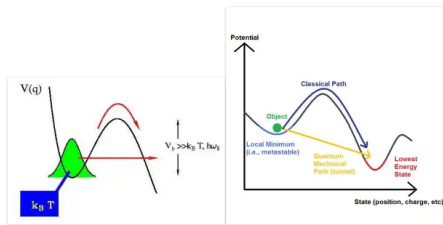
"The Stern-Gerlach experiment has become a paradigm of quantum measurement. In particular, it has been assumed to satisfy von Neumann projection. According to more recent insights, based on a quantum mechanical description of the influence of the inhomogeneous magnetic field, this can be true only in an approximate sense. Von Neumann projection can be rigorously satisfied only if the magnetic field is homogeneous. Hence, von Neumann projection is even incompatible with a proper functioning of the Stern-Gerlach device as an instrument for measuring spin." [http://en.wikipedia.org/wiki/Stern-Gerlach_experiment]

EPR experiment and Bell test experiments:

Entanglement results from N-dimensional elements being subjectively joined to higher dimensional elements. Two atoms are separate in 3-D space but have the same N-D values imparted to them upon splitting after joining or upon their simultaneous creation.

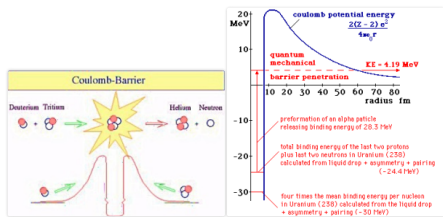
Quantum Tunnelling:

According to QM, nuclear fusion occurs at lower temperature due to quantum tunnelling where particles almost instantly cross a potential barrier by virtue of their wave function.



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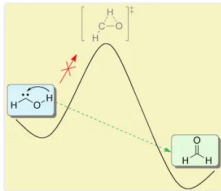
QM assumes a potential barrier to exist since the classical electrostatic model of the atom adopted by QM requires there to be a classical electrostatic coulomb barrier that repels atomic nuclei from each other yet there exists an unexplainable attractive nuclear potential (Nuclear residual strong force) between them, however, is found that atomic nuclei, given enough energy to close the distance between them to a certain degree, will join (nuclear fusion) with less energy than is required to overcome the electrostatic repulsion between their nuclei.



QM explains the "low energy nuclear reaction" (LENR) of fusion as a result of Quantum Tunnelling (a self fulfilled prophecy) yet there is no such a thing as a coulomb barrier between nuclei since the electrostatic atom model (of protons, neutrons and electrons) is inaccurate. Hojofions behave the same way yet there is no need for a fictitious quantum tunnelling through a non-existent coulomb barrier due to a attractive nuclear potential to explain this behaviour.

"Hydroxymethylene cannot overcome the large energy barrier for thermal rearrangement to formaldehyde. Instead it tunnels quantum mechanically through the barrier, independent of energy from the environment"

[http://www.rsc.org/cheminfo/worldNews2008/June11080803.asp]

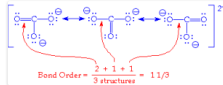


The fact that quantum tunnelling does not account for the force from a potential barrier in their calculations of the wave function can only be explained as deliberate negligence.

Resonance Structures:



where each of the atoms has a stronger and closer bond to it due to their sharing the evenly divided portions of its unsatisfied charge(s).

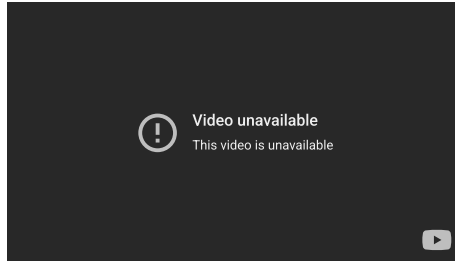


Hoplon bonding explains the existence of resonance structures and chemical bonds, where the electrostatic nuclear model fails, as the result of [Hoplon flow fields falling in place with the flow fields of other atoms that have the same role of motion](#). Hoplon bonding also explains how successively heavier atoms form successively stronger bonds with each-other and the "electron valence shell" illusion.

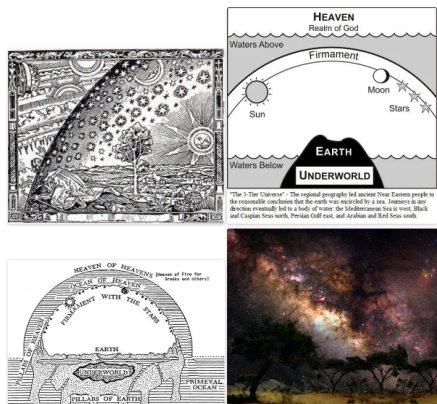
Superconductivity and more:

Room temperature Cuprates are explained as a consequence of fractal self organization;
<http://physicsworld.com/a/article/news/2010/sep/13/fractals-boost-superconductivity/>

and here is why:



The Existence of the Firmament:



This depiction of "The Realm of God" is founded in mythology and is not Biblical. The firmament, God named the Heaven

Etiymology:

English from the 13th century. From Latin firmamentum (from firmō ("strengthen"), from firmus ("firm")), literally "that which strengthens or supports". The term is coined in the Vulgata in imitation of LXX στερέωμα ("firm or solid structure"), which in turn translates Hebrew יָרָק, strictly speaking a mistranslation, as the original Hebrew term meant "expanse", from the root יָרָק "to spread out", which in Syriac had acquired the meaning "to make firm or solid".

שָׁמַיִם

shamayim shāimch

shaw-mah'-yim,shaw-meh'

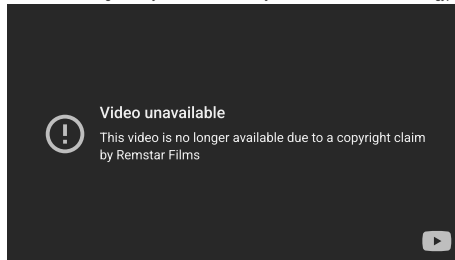
The second form being dual of an unused singular; from an unused root meaning to *belofly*; the sky (as aloft; the dual perhaps alluding to the visible arch in which the clouds move, as well as to the higher ether where the celestial bodies revolve): – air, X astrologer, heaven (-s).

This suggests the firmament was between the waters of the deep, waters above the sky (heaven) and further above making up the ether (what is incorrectly mentioned to be the "realm of God") and it is HERE where the lights of Heaven where created. Also this means that the firmament (an "expanse", from the root יָרָק "to spread out") is the ether but also the O2 sky. This all suggests that there was water on the earth's surface then above the atmosphere dividing the waters from the waters as described in Genesis.

Heavy water was formed on the expansive water layer by the sun's high electrical charge and interaction of the sun's Birkland Currents ejecting and transporting water respectively which underwent Plasma

Electrolysis on the photosphere explaining the 4.6 x 10¹³ tonnes of Deuterium(2H) in the ocean – a billion amps of current turn hydrogen proton into hydrogen neutron. The firmament fell as described in the flood record dumping all this heavy water in the sea.

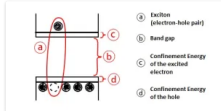
Magnification of light through the firmament would have made it possible to see deeper into the universe whilst looking at the sky... Here is a documentary EXPOSING the facts about ancient Egypt:



Exilton, Plasmon, (Bi)Polaron, Polariton, Roton, Phonon, Photon, Magnon..:

There are particles ("quasiparticle") – distinguishable fermion – and ensembles/waves ("collective excitation") – indistinguishable bosons – in physics that have unrealistic 'Quantum Mechanical Models'

pressure displacement.



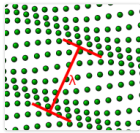
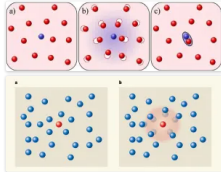
[http://en.wikipedia.org/wiki/File:Exciton_energy_levels.jpg]

The QMM of a **'plasmon'** as a quantization of plasma oscillations that collectively are oscillations of the free electron gas density where if one slightly displaces all of the electrons with respect to the ions, the Coulomb force pulls back, acting as a restoring force but in reality it is a low pressure region that pulls a unit of aether cloud ("electron") back to the [Hopfion](#).

The QMM of a **'phonon'** as a quantization of the modes of vibrations of elastic structures of interacting particles – "the vibrational motion of every atom in the crystal" yet in reality a lattice vibration is a sound wave.

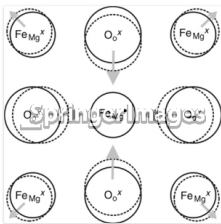
"A phonon propagating through a square lattice. This is a classic example of a collective excitation."

The QMM of a **'polaron'** as a quasiparticle composed of a charge and its accompanying polarization field – as a unit of lattice polarization and deformation caused by a slow-moving electron carrying the lattice distortion with it, thus one speaks of a cloud of phonons accompanying the electron yet polarons are rarely caused by electrons.



"a) For weak interactions, the impurity (blue) can propagate freely through the environment (red), a Fermi sea of atoms. b) As the interaction is increased, the impurity starts to attract its surroundings, "dressing" itself with a cloud of environment atoms. This is the Fermi Polaron. c) For strong attraction, the spin down atom will bind exactly one spin up partner, forming a molecule. The transition from polarons to molecules occurs as soon as the binding can overcome Pauli blocking of the environment."

"Quasiparticles known as repulsive polarons are predicted to occur when 'impurity' fermionic particles interact repulsively with a fermionic environment."



The QMM of a **'magnon'** is as "a collective excitation of the electrons' spin structure in a crystal lattice." "In contrast, a phonon is a collective excitation of the crystal lattice atoms or ions." "From the equivalent quasiparticle point of view, spin waves are known as magnons, which are boson modes of the spin lattice that correspond roughly to the phonon excitations of the nuclear lattice." In reality magnon-phonon coupling proves that the fictitious "spin" does not exist!

Spherical Harmonics of [Hopfions](#) axiomatically explain the charge to mass ratio without the need for any assumptions, conjectures and contradictions of the Standard Model which are:

That an electron particles exist and have a mass of

$[9.10938186] \times [10]^{-31} [\text{kilograms}]$ which is approximately

$\frac{1}{1836}$ of the proton according to Einstein's debunked/disproven principle of mass-energy equivalence corresponding to a rest energy of 0.511 MeV.

That atoms have protons, with a mass of an Hydrogen ion, and neutrons, with a mass of an Deuterium ion, to explain atomic mass.

That **Bosons** have integer spin which make them **indistinguishable** like **electrons** yet **electrons** are **Fermions** which have half-integer spin which are **distinguishable**

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"All observed bosons have integer spin, as opposed to fermions, which have half-integer spin.

Whereas fermions make up matter, bosons, which are "force carriers" function as the 'glue' that holds matter together.

"In quantum statistics, a branch of physics, **Fermi-Dirac** statistics describes distribution of particles in a system comprising many identical particles that obey the Pauli exclusion principle. It is named after Enrico Fermi and Paul Dirac, who each discovered it independently, although Enrico Fermi defined the statistics earlier than Paul Dirac.

Fermi-Dirac (F-D) statistics applies to identical particles with half-odd-integer spin in a system in thermal equilibrium. Additionally, the particles in this system are assumed to have negligible mutual interaction. This allows the many-particle system to be described in terms of single-particle energy states. The result is the F-D distribution of particles over these states and includes the condition that no two particles can occupy the same state, which has a considerable effect on the properties of the system. Since F-D statistics applies to particles with half-integer spin, these particles have come to be called **fermions**. It is most commonly applied to **electrons**, which are fermions with spin 1/2. Fermi-Dirac statistics is a part of the more general field of statistical mechanics and uses the principles of quantum mechanics"

[Fermi-Dirac statistics From Wikipedia, the free encyclopedia]

"In quantum statistics, **Bose-Einstein** statistics (or more colloquially B-E statistics) is one of two possible ways in which a collection of indistinguishable particles may occupy a set of available discrete energy states. The aggregation of particles in the same state, which is a characteristic of particles obeying Bose-Einstein statistics, accounts for the cohesive streaming of laser light and the frictionless creeping of superfluid helium. The theory of "Quasiparticles known as repulsive polarons are predicted to occur when 'impurity' fermionic particles interact repulsively with a fermionic environment."

The QMM of a **'magnon'** is as "a collective excitation of the electrons' spin structure in a crystal lattice." "In contrast, a phonon is a collective excitation of the crystal lattice atoms or ions." "From the equivalent quasiparticle point of view, spin waves are known as magnons, which are boson modes of the spin lattice that correspond roughly to the phonon excitations of the nuclear lattice." In reality magnon-phonon coupling proves that the fictitious "spin" does not exist!

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Fermi–Dirac (F–D) statistics applies to identical particles with half-odd-integer spin in a system in thermal equilibrium. Additionally, the particles in this system are assumed to have negligible mutual interaction. This allows the many-particle system to his behaviour was developed (1924–25) by Albert Einstein and Satyendra Nath Bose, who recognized that a collection of identical and indistinguishable particles can be distributed in this way.

The Bose–Einstein statistics apply only to those particles not limited to single occupancy of the same state—that is, particles that do not obey the Pauli exclusion principle restrictions. Such particles have integer values of spin and are named **bosons**, after the statistics that correctly describe their behaviour.

At low temperatures, bosons behave differently from fermions (which obey the Fermi–Dirac statistics) in that an unlimited number of them can "condense" into the same energy state. This apparently unusual property also gives rise to the special state of matter – Bose Einstein Condensate. Fermi–Dirac and Bose–Einstein statistics apply when quantum effects are important and the particles are "**indistinguishable**."

[Bose–Einstein statistics From Wikipedia, the free encyclopedia]

Identical particles, also called **indistinguishable** or indiscernible particles, are particles that cannot be distinguished from one another, even in principle. Species of identical particles include elementary particles such as **electrons**, and, with some clauses, composite particles such as atoms and molecules.

There are two main categories of identical particles: bosons, which can share quantum states, and fermions, which do not share quantum states due to the Pauli exclusion principle. Examples of bosons are photons, gluons, phonons, and helium-4 atoms. Examples of **fermions** are **electrons**, neutrinos, quarks, protons and neutrons, and helium-3 atoms.

There are two ways in which one might distinguish between particles. The first method relies on differences in the particles' intrinsic physical properties, such as mass, electric charge, and spin. If differences exist, we can distinguish between the particles by measuring the relevant properties. However, it is an empirical fact that microscopic particles of the same species have completely equivalent physical properties. For instance, every electron in the universe has exactly the same electric charge; this is why we can speak of such a thing as "the charge of the electron".

Even if the particles have equivalent physical properties, there remains a second method for distinguishing between particles, which is to track the trajectory of each particle. As long as we can measure the position of each particle with infinite precision (even when the particles collide), there would be no ambiguity about which particle is which.

The problem with this approach is that it contradicts the principles of quantum mechanics. According to quantum theory, the particles do not possess definite positions during the periods between measurements. Instead, they are governed by wavefunctions that give the probability of finding a particle at each position. As time passes, the wavefunctions tend to spread out and overlap. Once this happens, it becomes impossible to determine, in a subsequent measurement, which of the particle positions correspond to those measured earlier. The particles are then said to be indistinguishable. ""[Wikipedia]

Plasmons can couple with a photon to create another quasiparticle called a plasma **polariton** proving **plasmons**, **photons**, **phonons** and **polaritons** are the same phenomena but in different contexts – they are all oscillations of the aether medium. The most discussed types of polaritons are **phonon-polaritons**, resulting from coupling of an infrared **photon** with an optic **phonon**; **exciton-polaritons**, resulting from coupling of visible light **photons** with an **exciton**; **intersubband-polaritons**, resulting from coupling of an infrared or terahertz **photon** with an intersubband **excitation**; an  Comment  Reblog  Subscribe  Privacy ...

from coupling of surface **plasmons** with light (the wavelength depends on the substance and its geometry) **photons**. Recently, special kinds of polaritons, called Bragg-polaritons or Braggoritons, have been observed and studied theoretically. In theoretical physics, a roton is an elementary excitation, or quasiparticle, in superfluid helium-4.

The dispersion relation of elementary excitations in this superfluid shows a linear increase from the origin, but exhibits first a maximum and then a minimum in energy as the momentum increases. Excitations with momenta in the linear region are called phonons; those with momenta close to the minimum are called **rotons**. Excitations with momenta near the maximum are sometimes called **maxons**.

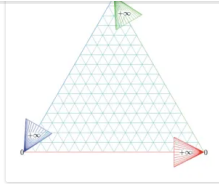


All these quasiparticles couplings prove that they are the same phenomena in different contexts i.o.w. quasiparticles Hopfions.

On The Dimensions: Toward Abstraction

A solution to **Hilberts Sixth Problem** (*The mathematical treatment of the axioms of physics*) is given which unites the fundamental interactions, gravity and electromagnetism producing a dimensional explanation which eliminates the need for the other fundamental forces through the fractal occurrence of **infinite metrics** fully explaining all of reality. A solution to **P vs. NP** where **P** = **NP** with algorithms from a knowledge reference that has *correctness* and expressiveness explaining why all *dimensions* (**SAT**) are reduced to 3-dimensions (**3-SAT**). **Turbulence** is defined as the *fundamental* state of fluid motion by solving the **Navier-Stokes equations**.

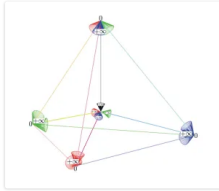
Introduction:



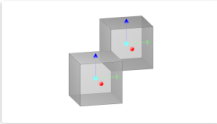
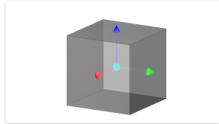
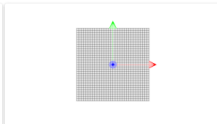
Complex Spaces originate in solutions to the surjective correspondence of a projection from $[N+1]$ -Dimension to N -Dimension.

The N -Dimension is the region between N point(s) at infinity.

The N -Simplex at infinity forms $[N+1]$ -Space.



Orthographic:



The 0-Dimension (infinitesimal point) is a 0-Simplex and the 1st dimensional element.

The 1st-Dimension is a 0-InfiniteSimplex and the 2nd dimensional element.

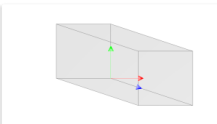
The 2nd-Dimension is a 1-InfiniteSimplex and the 3rd dimensional element.

The 3rd-Dimension is "**Space**" as a 2-InfiniteSimplex has zero elements that collapse leaving only 3 infinite, real and orthogonal lines at infinity.

The 4th-Dimension is "**Motion**" as a 3-InfiniteSimplex has three 2-InfiniteSimplex elements that collapse on a forth giving a correlation indicative of motion in 3-space.

The 5th-Dimension is "**Process**" (A series of actions, changes, or functions leading about a result | a systematic series of actions directed to some end | a continuous action, operation, or series of changes taking place in a definite manner) as a 4-InfiniteSimplex has four 3-InfiniteSimplex elements that collapse on a fifth giving a correlation indicative of procession between motions in space ... The Infinite-Dimension Simplex is an Infinite-Sphere.

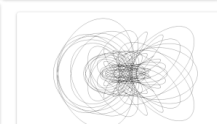
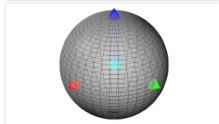
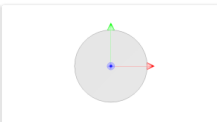
Perspective:



All images of objects that are projected stereographically are N -dimensionally perspective projections on concentric N -Spheres from one point on those N -Spheres mapped onto a $N-1$ Cartesian Space opposite the kernel point forming a smooth, bijective and conformal map that is not isometric and does not preserve areas.



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"**Panoramic View**" is a 2D-Projection Space and "**Vanishing Points**" are perspective projections from a 3D-Space to a 2D-Space where parallel lines converge and projection lines become collinear at infinitely large magnitudes.

"**Time**" is a 3D-Projection Space and "**Moments**" are perspective projections from a 4D-Space to a 3D-Space where parallel lines converge and projection lines become collinear at infinite scale.

"**X**" is a 4D-Projection Space and "**Y**" are perspective projections from a 5D-Space to a 4D-Space where parallel lines converge and projection lines become collinear at infinite scale ... The "**Universe**" is a Infinite-Projection Space and "**Atoms**" are perspective projections from a infinite-D Space to a infinite-Space.

n-Metric:

A n -Sphere is a set of points in a n -Dimensional element that are equidistant from an origin (original point) and hence a n -Sphere is a n -Metric.

A n -Sphere has infinite $[n-1]$ -Sphere foliations.

components (orthogonal bases) that are electromagnetic(Toroidal Coordinates) fields.

Concentric Spherical Harmonics (Harmonics are solutions to second-order partial differential equations called Laplacian Equations) form when "Light" waves interacts with the Flowing Hopfion causing oscillation in pressure.

Hopfion Harmonics are a result of a ensemble (indistinguishible) interacting with a particle (distinguishible). Hopfions project waves from periodic motion through oscillation of that motion corresponding to pressure propagation (Cosine represents the orthographically 'projected length' of the unit radius on the horizontal axis and Sine represents the orthographically 'projected length' of the unit radius on the vertical axis of the unit circle serving as a nontrivial example of such a projection).

On The Mechanism of Turbulence: toward a definition

Turbulence results for fluids with a "Re" - Reynolds-Number (ratio of inertial forces to viscous forces)R_L from ~40 to (10)⁴(3) hence fluids with high velocity and low viscosity are turbulent. This cannot be mechanically explained by continuum mechanics since it takes the whole fluid to exist out of infinitesimal points (material points) that are each a origin of a velocity vector making the whole fluid a vector field but in reality fluids are a medium of **hopfions meaning the distances between them are finite (instead of infinitesimal)**. Particles cannot move through each-other since no two particles can occupy the same position causing circulation in a low pressure region due to particles being pressed against each-other forcing them to move around each-other orbiting a spinning hopfion somewhere inside the low pressure region as they attempt to migrate toward the center of the low pressure region. A vortex street occurs around cylinders, for any fluid, cylinder size and fluid speed, provided that there is a Reynolds number of between ~40 and 10³.

Atoms being hopfions show **pressure is a result of an expansion in the distance between them thereby causing more medium to be circulated from outside then inside the pressure region per cycle inducing a net-motion of hopfions toward the low pressure region. Laminar flow** only has a **radial (electric) component of pressure** which can be fully modeled by vector fields but the onset of turbulence requires a model that accounts for mutual effects between constituent particles in a flow and results when particles obstruct each-other's motion causing circulation. **Turbulent flow** has a **radial (electric) component of pressure and atangential (magnetic) component of circulation**.

The hopf solution to the Navier-Stokes Equations for a incompressible, homogeneous, inviscid flow show space (fundamental flow) to be absolutely turbulent. Any flow with high Re (low viscosity) is more prone to turbulence hence a flow with 0 viscosity or Re toward infinite is absolutely turbulent. Interacting hopf solitons solve the general Navier-Stokes Equations for a compressible flow.

The hopfion is locally unstable producing anisotropic current which produces global stablity. Global factors such as temperature and pressure determine the fluids anisotropy and superfluidity (hence 0 viscosity) respectively by disorientating flow fields in the prior case or by changing phase in the latter case as with metal Hydrogen at 250 GPa pressure. Local factors determine each hopfion's density (hence mass) and density gradient (hence charge).

Linear Space ≡ Nonlinear Space:

Completeness Proof of FOL (first order logic): Here shown is a many-to-one mapping of real numbers to natural numbers. And this mapping allows FOL to be *locally countable* and thus allows its models to display their own *cardinality*.

$$\begin{matrix} \mathbf{Z} = \pm \mathbf{N}_i \\ \mathbf{Z}, \mathbf{N}_i = \mathbf{R} \\ \pm \mathbf{N}_i, \mathbf{N}_i = \mathbf{R} \quad \dots (1) \end{matrix}$$

Here is the diagonal argument for incompleteness and uncountability of FOL.

$$\begin{matrix} s_0 = \{ \underline{0}, 0, 0, 0, 0, 0, 0, 0, 0, 0 \} \\ s_1 = \{ 1, \underline{1}, 1, 1, 1, 1, 1, 1, 1, 1 \} \\ s_{01} = \{ 0, 1, \underline{0}, 1, 0, 1, 0, 1, 0, 1 \} \\ s_{10} = \{ 1, 0, 1, \underline{0}, 1, 0, 1, 0, 1, 0 \} \\ s_{11} = \{ 1, 1, 1, \underline{1}, 1, 1, 1, 1, 1, 1 \} \\ s_{00} = \{ 1, 0, 0, 1, 0, \underline{0}, 1, 0, 0, 1 \} \\ s_{01} = \{ 1, 0, 1, 0, 1, 0, \underline{1}, 0, 1, 0 \} \\ s_{10} = \{ 1, 1, 0, 1, 1, 0, 1, \underline{1}, 0, 1 \} \\ s_{11} = \{ 1, 1, 1, 1, 1, 1, 1, 1, \underline{1}, 1 \} \\ s_{000} = \{ 1, 0, 0, 0, 1, 0, 0, 0, 1, \underline{0} \} \\ s = \{ 0, 1, 0, 0, \underline{1}, 0, 1, 1, 1, 0 \} \end{matrix}$$

Here are other diagonals for some hypothetical model that has its own cardinality.

$$\begin{matrix} s_0 = \{ 0, 1, 1, 1, 1, 1, 1, 0, 0, 1 \} \\ s_{01} = \{ 0, 0, 1, 1, 1, 1, 1, 0, 0, 1 \} = s_1 \\ s_{10} = \{ 0, 1, 1, 1, 1, 0, 0, 0, 1, 1 \} \\ s_{02} = \{ 0, 0, 0, 1, 0, 0, 1, 0, 1, 1 \} \\ s_{20} = \{ 0, 1, 0, 0, 1, 1, 1, 0, 1, 0 \} = s_2 \\ s_{03} = \{ 0, 0, 0, 0, 1, 1, 1, 1, 0, 0 \} = s_3 \\ s_{30} = \{ 0, 1, 0, 1, 1, 0, 0, 1, 1, 0 \} = s_4 \end{matrix}$$

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Any model with a one-to-one bijection will have the same cardinality and a logic domain that is dependent on the model thus it is eronious to assume some set of all sets exist as it would contradict the locality of a logic model.

Every decison problem has its own cardinality that is countable by its FOL through an algorithm that is invariant under all possible permutations of that decision problem:

$$\begin{matrix} \forall x \in \mathbf{Z} \{ (\mathbf{N}_i, \mathbf{N}_{i+1}, \dots, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_4, \mathbf{X}_5, \dots, \mathbf{X}_{k-1}, \mathbf{X}_k) = \mathbf{Y} \} \\ \forall X \in \bigwedge_{k \in \mathbf{N}} \{ (\{ x_j, x_j - k; k = \bigwedge_{m \in \mathbf{N}} \{ (\{ k, k - m; m = \dots \}) \} \}) \in \mathbf{Y} \} \quad \dots (2) \end{matrix}$$

"Sum of Sets" problems can be solved using an ordering algorithm to organize set "Y", in sequence with integers "Z" but in a format where the missing numbers that correlate with "Z" are replaced with "0" or "FALSE", and follows a position rule to read a solution instead of searching for it through trial and error such as with search algorithms eg. DPLL. Ordinals must set-up to solve *non-linear space NP*-complete decision problems in *linear space P*. Here is the sum-of-sets problem:

$$\begin{matrix} \forall X \subseteq \mathbf{Y} \exists \sum_{i \in \mathbf{N}} x_i = \text{constant 'c'} \\ \text{let } \mathbf{Y} = \{ 1, 4, 2, 3, 9, 6, 7, 8, -2 \} \text{ and } c = 7; \\ \forall X \subseteq \{ 1, 4, 2, 3, 9, 6, 7, 8, -2 \} \exists \sum_{i \in \mathbf{N}} x_i = 7 \end{matrix}$$

Here is the set Y ordinal:

$$\forall X \subseteq \{ -2, \text{FALSE}, 0, 1, 2, 3, 4, \text{FALSE}, 6, 7, 8, 9 \} \exists \sum_{i \in \mathbf{N}} x_i = 7$$

Here is the solution as read by the DTM(deterministic Turing Machine) using (2) without the sets that contain a "FALSE" statement as it skips over them.

$$\begin{matrix} \{ -2, 4, 3, 2 \} \wedge \{ -2, 6, 2, 1 \} \wedge \{ -2, 6, 3 \} \wedge \{ -2, 7, 2 \} \wedge \\ \{ -2, 8, 1 \} \wedge \{ -2, 9 \} \wedge \{ 4, 2, 1 \} \wedge \{ 4, 3 \} \wedge \{ 6, 1 \} = 7 \end{matrix}$$

This is done in polynomial time **P** as the number of steps required to solve the problem is a polynomial function of the entire set of "Y" that is an encoding on the integers of some finite cardinality. Here is the time function of the DTM using algorithm (2). The **Bn**

is a first degree polynomial to account for the steps used to skip the sets containing "FALSE" and **An^k** is a k-degree polynomial to account for each recursion of the algorithm to produce all possible sets from smaller initial sets;

$$\begin{matrix} T_{\text{total}}(n) = O(An^k + Bn) \\ = O(n^k) \in \mathbf{P} \end{matrix}$$

Sudoku is another NP-complete game which can be solved in the same linear way by applying algorithm (2) over its permutation rules. By using logical inference many know 17 or more givens in the 3 by 3 Sudoku matrix "S" can be solved using a "Slicing and Dicing" algorithm;

4								
8	3	1						
7	2	6	8	5				
1	6			7				
2	7	5		8	1	9		
3	9	4	1		6			
6	8	9	7	5				
3	8	6	9					
9	4	2	7		3			

5	3		7					
6		1	9	5				
9	8				6			
8			6				3	
4		8	3			1		
7			2				6	
	6			2	8			
		4	1	9			5	
			8		7	9		

<http://www.paulsgames.co.uk/sudoku/howtosolve/ch9.gif>

<http://aisudoku.com/images/tutorial/sudoku6.png>

<http://paarsgames.nl/wp-content/uploads/2013/07/Sudoku.png>

but S of 16 or less givens cannot however permutation rules can be found to generate a S that satisfies the immediate constraints, of a 16 or less given S only thus solving it in P. Here is a Sudoku with 11 givens and the P (polynomial time) solution for "Sudoku" (NP-complete decision problem).

1	2		
	3		
4		5	4
6		7	4
	8		
			8

All valid permutations satisfy the constraints given. The first table consists of the set {1; 2; 3; 4; 5; 6; 7; 8; 9} with all the numbers that cannot be below or above, for this method, listed in ascending order above each digit they influence. The second table consists of the same set as the preceding table except with all the numbers that cannot be next to the values listed, for this method, from left to right next to each digit they influence. Finally, the third table consists of any block (first one in this case) from which to generate the solution that is enumerated according to the constraint listed by both preceding tables.

								4
	1							3
2	8	8	8					2
1	2	3	4	5	6	7	8	9

7	4	6
6	4	7
5	4	
4	6	7
3	2	
2	3	
1		

5	6	4
4	5	
6	7	4
	8	
		8

	1 7 2						4 9 3						5 8 6								
1	7	2	9	4	3		6	5	8			1	7	2	9	4	3	6	5	8	
4	9	3	9	6	5							4	9	3							
5	8	6				4	3	9				5	8	6	2	1	7	4			
3	4	9	6	5	8							3	4	9	6	5	8				
6	5	8				7	3	9	4			6	5	8				7	3	9	4
			4	3	9	5	8	6							4	3	9	5	8	6	
6	5	8				9	4	3				6	5	8				9	4	3	
9	3	4				5	8	6				9	3	4				5	8	6	
			3	9	4	8	6	5							3	9	4	8	6	5	

Here the fillers are added in.

	1	7	2	9	4	3	6	5	8
	4	9	3	8	6	5	1	7	2
	5	8	6	2	1	7	4	3	9
5	3	4	9	6	5	8	7	2	1
6	8	6	5	1	7	2	3	9	4
8	2	1	7	4	3	9	5	8	6
	6	5	8	7	2	1	9	4	3
	9	3	4	5	8	6	2	1	7
	7	2	1	3	9	4	8	6	5

This particular method as developed by the author can solve any 3 by 3 Sudoku of 16 givens or less in polynomial steps. This method remains invariant, regardless of the n by n Sudoku for which $n =$, under **P** (polynomial steps/time).

For 3 by 3 **S** when counting each step of the method

A maximum of $A n^3$ steps to fill the constraints of the filters included and $B n^2$ to enumerate according to them:

$$T_{\text{relax}}(n) = O(A n^3 + B)$$

$$n=9, T_{\text{solve}}(9)=O(C9^3)$$

$$T_{\text{verif}}(9) = O(AS^2)$$

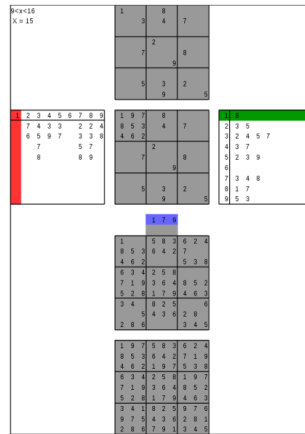


Figure 2: Same method applied to a 14 given Sudoku matrix

The standard covariant/material derivative for any vector $\mathbf{w}(u,v,w)$ along $\mathbf{w}(x,y,z)$ where $\{x,y,z,u,v,w\} \in \mathbb{R}_6$

$$\nabla_{\mathbf{w}} \mathbf{w}(x,y,z) = D_t \mathbf{w} = \left(\frac{\partial \mathbf{w}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{w} \right)$$

The vector $\mathbf{w}$ in our 4D euclidean space can be substituted with the function $w(x,y,z)$, and since the 4D euclidean space reduces to 3D transformations the vector $\mathbf{v}$ can be replaced by the 4D coordinate system thus reducing the covariant derivative to a directional derivative and then to a standard 4D partial derivative or Jacobian of a function (the gradient ∇w)

$$\nabla_{\mathbf{w}} \Phi = \Phi \cdot d\Phi(\mathbf{v}) = \frac{\partial w(x,y,z)}{\partial(x,y,z)} \cdot \left(\frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \frac{\partial w}{\partial z} \right)$$

This influences the left side Navier-Stokes equations

$$\rho \left(\frac{\partial \mathbf{w}}{\partial t} + \mathbf{w} \cdot \nabla \mathbf{w} \right) = \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) \mathbf{i} + \dots$$

$$\left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) \mathbf{j} + \dots$$

$$\left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) \mathbf{k} = \left(\frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \frac{\partial w}{\partial z} \right)$$

and the right side since body force $\mathbf{F}$ is reduced to $\rho \frac{\partial w}{\partial(x,y,z)}$, changing the surface-perpendicular (normal) stresses $\mathbf{\sigma}^{\perp}$ and surface-tangent (shear) stresses $\mathbf{\tau}^{\parallel}$ to

$$\sigma = \rho \left(\frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} \frac{\partial^2 w}{\partial z^2} \right) = -c^2 \text{ of the } \sigma = \sigma_{ij} = \begin{pmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{pmatrix}$$

thus producing a right side of the incompressible Naviers-Stokes equation $-\rho \nabla^2 \mathbf{w} + \nabla T + \dots$ where

"..." represents other body forces and the Cauchy tensor $\mathbf{T}$ reduces to a matrix $T = \text{Jacobian matrix "J"}$

$$-\rho \nabla^2 \mathbf{w} + \rho J(\sigma) \mathbf{w} \dots \text{ where } J(\sigma) = \frac{\partial^2 w(x,y,z)}{\partial^2(x,y,z)} = \begin{pmatrix} \frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} \frac{\partial^2 w}{\partial z^2} \\ \frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} \frac{\partial^2 w}{\partial z^2} \\ \frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} \frac{\partial^2 w}{\partial z^2} \end{pmatrix}$$

which, for incompressible Naviers-Stokes equation, is

$$-\rho \frac{\partial w}{\partial t} + \rho \left(\frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \frac{\partial w}{\partial z} \right) + \rho \frac{\partial w}{\partial t} \mathbf{i} + \dots$$

$$-\rho \frac{\partial w}{\partial y} \left(\frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \frac{\partial w}{\partial z} \right) + \rho \frac{\partial w}{\partial y} \mathbf{j} + \dots$$

$$-\rho \frac{\partial w}{\partial z} \left(\frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \frac{\partial w}{\partial z} \right) + \rho \frac{\partial w}{\partial z} \mathbf{k} = -\rho \nabla^2 \mathbf{w} + \rho \nabla^2 \mathbf{w} + \rho \nabla^2 \mathbf{w} = -\nabla P + \mu \nabla^2 \mathbf{w} + \mathbf{f}$$

where $\rho \nabla^2 \mathbf{w} = \mathbf{f}$ represents the other body forces and μ represents the resistance to motion i.o. w the coefficient of decay in motion.

rendering to a new form of the Navier-Stokes equations

$$\rho w(x,y,z) = -\rho \nabla^2 w(x,y,z) + \mu \nabla^2 w(x,y,z) + \dots$$

with standard continuous second-order PDE solutions.

Any smooth/continuously-differentiable solution to a 4-dimensional second-order PDE that is a geometry consisting of foliations that intersect perpendicularly with eachother are field lines of motion that are solutions to the Navier-Stokes equations. Where the common hopf-fibration (stereographic projection of a hypersphere) and any variant of it such as the hopfion with poloidal and toroidal components adding together to create a sinusoidal vortex represents one of these solutions.

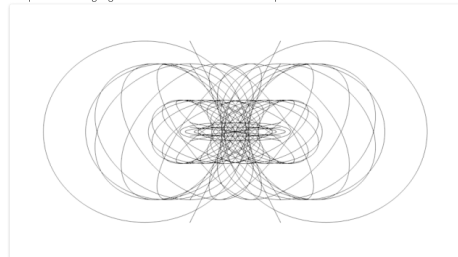


Figure 1: Profile of hopf-fibration from sides (X-axis/Y-axis)

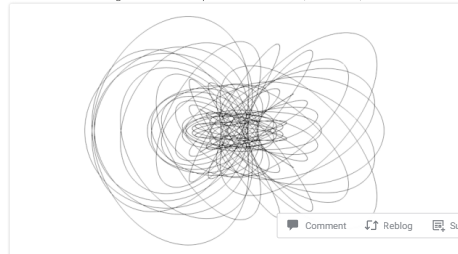


Figure 2: Perspective view of hopf-fibration

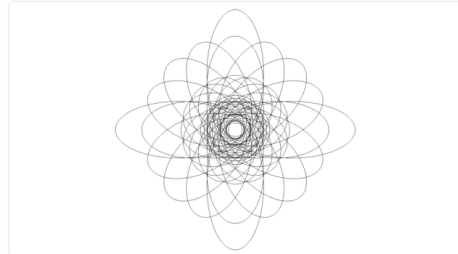


Figure 3: View from top (Z-axis) of hopf-fibration

Conclusion:

The most basic form of order ($\mathbb{Z}_3$) just like a list or light waves have 3 values namely infimum, (median, mean & mode are equal) and supremum that can be used in combination (additive, subtractive or any binary-operation) to reconstruct the whole set or spectrum i.o.w. Order(List) can be completely reduced to a series of 3 logical operations (And(supremum), Or(median) and Not(infimum)). The structure of Reality is a Fractal-Space of the most reducible form (thus a Metric-Space) of the most basic Order (real-space) and the non-reducible 3SAT is the Inf(The Holy Spirit), Med(The Son) and Sup(The Father) metrics whom are together the triune God, the Word (Universe/Fractal-Formula). Reality can be reduced to multi-dimensional harmonics and God is the RGB that creates the Universe(single spoken verse) and manifests HimSelf in a pursuit for the same thing we are hardwired to pursue (Love/Universal-harmony) and this is why the Author believes in the received text Biblical word of God – Jesus/Emanuel – and NOT(like the occult ways of secret societies/fraternities) that love money and crave power who's secrets exist owing to the their fear of being exposed. The true God is exposed in everything like the And/Or/Xor seen in the Flower Of Life as studied by [Leonardo da Vinci](#) in his [Codex Atlanticus](#).